Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Close the fuel supply valve. See equipment manufacturer service information.
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-018-tr-gc.html)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/39/39-002-

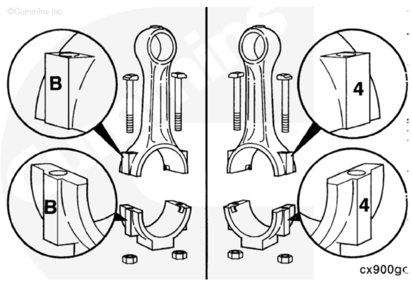
004-tr-gc.html)
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-037.html)
- Remove the lubricating oil pan and gasket. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-025-tr-gc.html)
- Remove lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-035-tr-gc.html)
- Remove and disassemble the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/39/39-001-054-tr.html)

Clean and Inspect for Reuse

The letter or number on the connecting rod cap **must** be the same as the letter or number on the connecting rod.

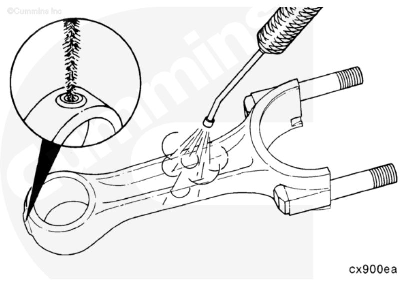
Do **not** assemble a new cap to a used connecting rod, or a used cap to a new connecting rod.

Remove the connecting rod capscrew nuts and caps from the connecting rods.



⚠ WARNING ⚠

When using a steam cleaner, wear protective clothing and safety glasses or a face shield. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam to clean the rods.

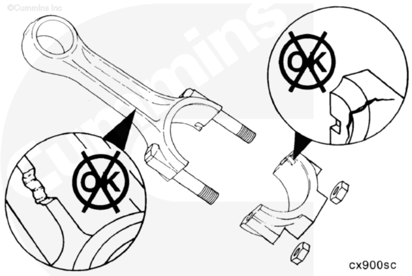
Use a soft bristle brush to clean the oil drilling.

Dry with compressed air.

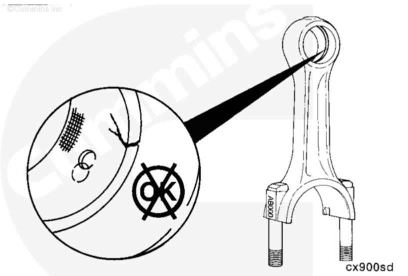
Inspect the connecting rod and cap for fretting damage on the mating surfaces.

The rod and cap **must** be replaced as an assembly if any fretting damage is visible on either piece.

Replace the connecting rod if the I-beam is nicked or damaged.

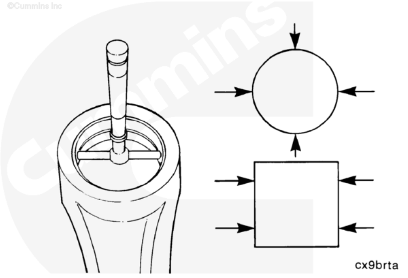


Inspect the connecting rod pin bore bushing for damage or misalignment of the oil passage and bushing.



Measure the connecting rod piston pin bushing inside diameter.

If the connecting rod piston pin bushings are **not** within the given specifications, the piston pin bushings can be replaced.

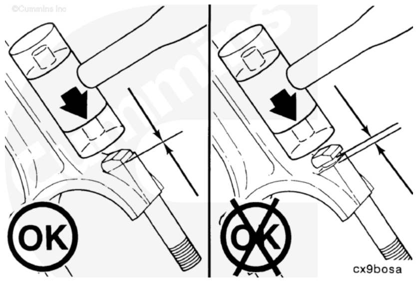


Connecting Rod Piston Pin Bore Diameter (Bushing Installed)		
mm		in
45.023	MIN	1.7726
45.060	MAX	1.7740

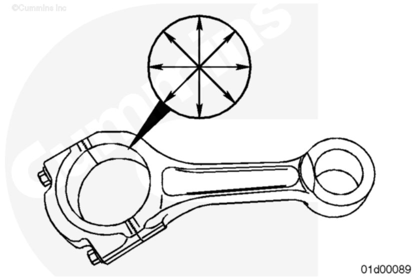
Note : Service replacement rod bushings are available. See the appropriate parts catalog for part numbers.

The bushing **must** be precision machined after installation. If machining capability is available, the bushing can be replaced.

Tap the connecting rod bolts in until the head is aligned and seated on the flat machined surface of the connecting rod.

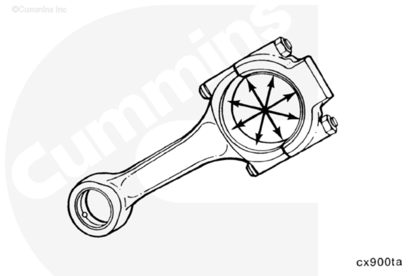


Measure the connecting rod crankshaft bore with the bearing shells removed and the caps tightened to the proper torque value. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/39/39-001-054-tr.html)



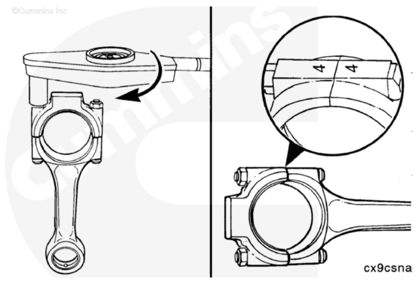
Measure the connecting rod crankshaft bore inside diameter.

Connecting Rod Crankshaft Bore I.D. (Without Bearing)		
mm		in
80.987	MIN	3.1885
81.013	MAX	3.1895



The connecting rod **must** be assembled with the capscrew nuts tightened to specifications before stamping the cylinder identification number on the connecting rod.

If a new connecting rod is installed, **always** stamp the new connecting rod with the cylinder number of the rod being replaced.



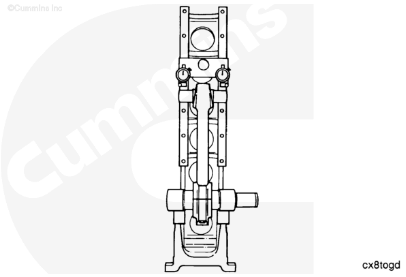
Bend and Twist Inspect

Calibration Procedure

Use a connecting rod checking fixture, Part Number ST-561, and a connecting rod mandrel set, Part Number 3823286, to inspect the bend and twist of the rods.

Calibrate the checking fixture with a new rod that has been measured for correct center to center length, 215.975 to 216.025 mm [8.5029 to 8.5040 in].

Assemble the connecting rod cap to the rod as described previously in this procedure.

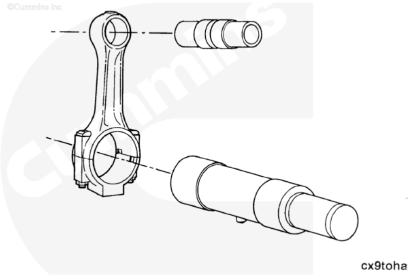


Install the piston pin mandrel from the connecting rod mandrel set, Part Number 3823286, into the piston pin bore.

Note : Use a mandrel, Part Number 3823283, if the piston bushing has been removed or mandrel, Part Number 3823284, if the bushing is still in place.

Install the mandrel, Part Number 3823303, into the crankshaft bore and expand the mandrel.

Note : Make sure the pin on the mandrel is pointed down and locked in position in the center of the connecting rod.

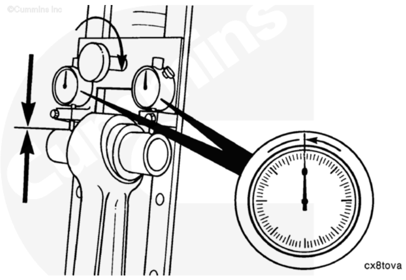


Install the connecting rod into the fixture.

Move the dial holder to position the contact points of the indicators on the mandrel in the piston pin bore.

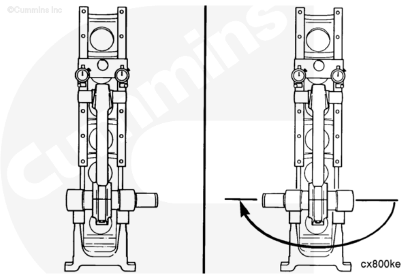
Tighten the bracket to hold the indicators in position.

Set the dial indicators to read zero.



Remove the connecting rod from the fixture.

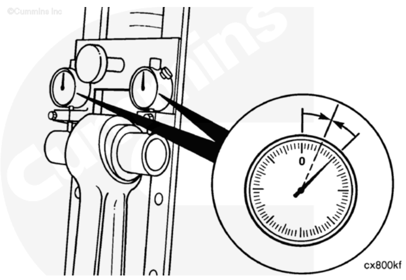
Turn the rod 180 degrees horizontally, and install the rod into the fixture again.



Check the dial indicators for the zero position again.

If the dial indicators show any change from zero, adjust the dials to half the indicated reading.

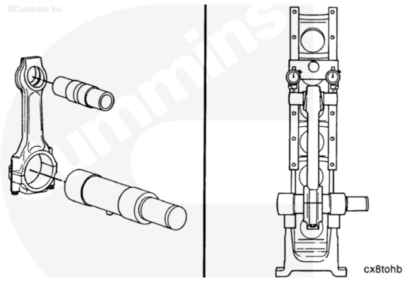
The fixture is now calibrated to allow the connecting rod to be installed into the fixture in either direction, and the dials will indicate an equal deflection on either side of zero.



Test

Install the mandrel and arbor into the connecting rod to be inspected.

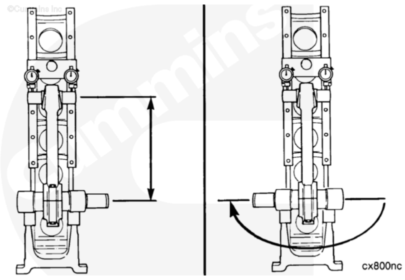
Install the connecting rod into the fixture.



Measure the connecting rod length and bend (alignment).

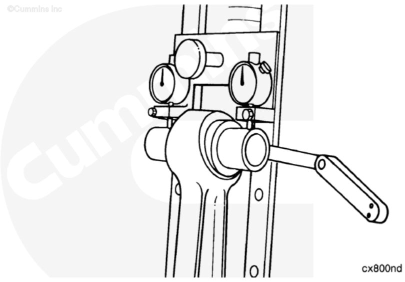
Connecting Rod Length		
mm		in
215.975	MIN	8.5029
216.025	MAX	8.5049

Connecting Rod Bend (Alignment)				
	mm		in	
Bushing removed		0.20	MAX	0.008
Bushing installed		0.15	MAX	0.006



Install a feeler gauge between the mandrel and the dial indicator holding plate as shown.

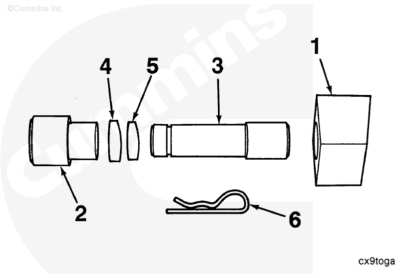
Connecting Rod Twist				
	mm		in	
Bushing Removed		0.50	MAX	0.020
Bushing Installed		0.30	MAX	0.012



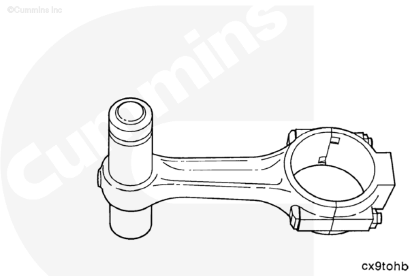
Replace

This tool is used to remove and install the bushings to the correct position.

Connecting Rod Pin Bore Bushing Removal/Installation Tool, Part Number 3823690			
Reference Number	**Part Number**	**Description**	**Quantity**
1	3823691	Anvil	1
2	3823693	Cup	1
3	3823692	Mandrel	1
4	3823694	Driver ring (thick)	1
5	3823695	Knock-out-ring (thin)	1
6		Hair pin cotter pin	1

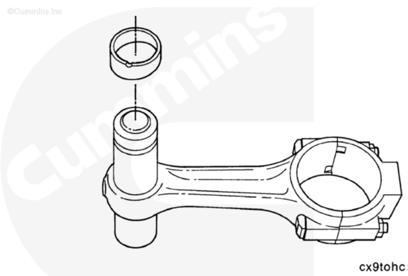


Place the pin bore end of the connecting rod on the mandrel.

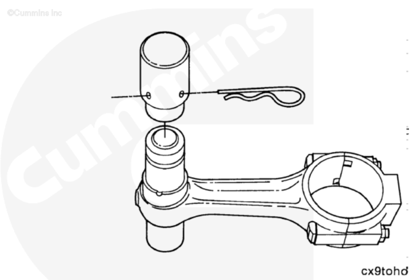


Place the knock-out (thin ring) on the mandrel on top of the connecting rod.

Match up the angle surfaces.



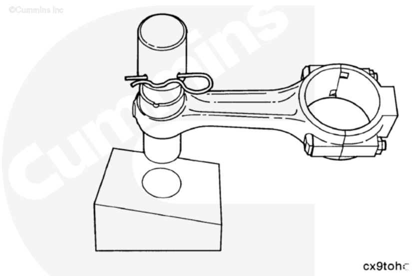
Install the cup on the mandrel and secure with the hair pin cotter pin.



Install the mandrel in the anvil.

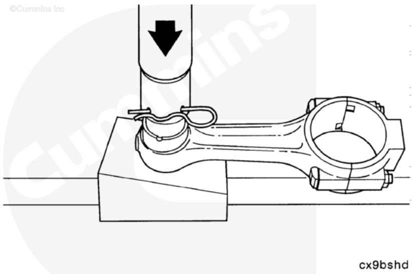
Place the connecting rod so the angle on the rod matches with the angle on the anvil.

The connecting rod will be in the horizontal position.


cx9tohc

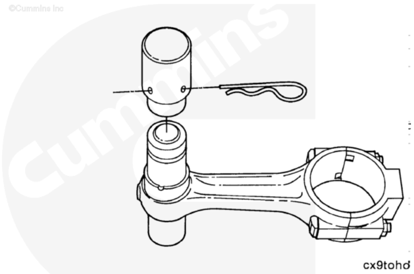
Press the connecting rod bushing out by applying force on top of the cup.

Press the bushing through the bore.


cx9bshd

Remove the hair pin cotter pin.

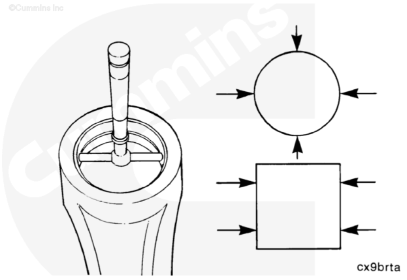
Remove the cup and slide the connecting rod from the mandrel.


cx9tohd

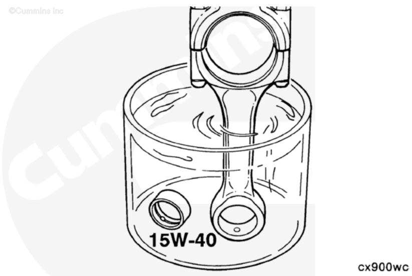
Measure the connecting rod bushing bore with the bushing removed.

Note : If the connecting rod bushing bore is not within specifications, replace the connecting rod.

Connecting Rod Piston Pin Bore Diameter (Bushing Removed)		
mm		in
48.988	MIN	1.9286
49.012	MAX	1.9296

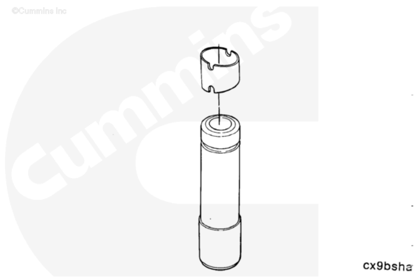

cx9brta

Before installing the connecting rod bushing, submerge the bushing and connecting rod pin end in clean 15W-40 engine oil.



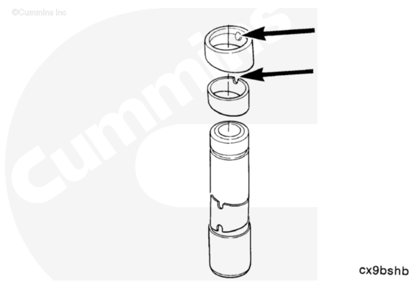
cx900wc

Place the connecting rod bushing on the mandrel.

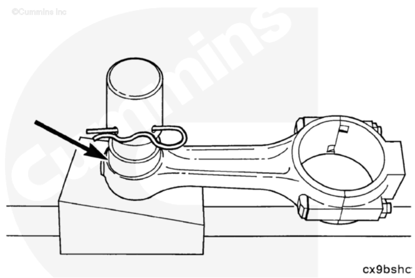


Place the knock-out (thin ring) inside the driver ring.

Match the angles, place the angle side down and align the notch of the knock-out (thin ring) to the pin in the driver ring and slide on the mandrel.



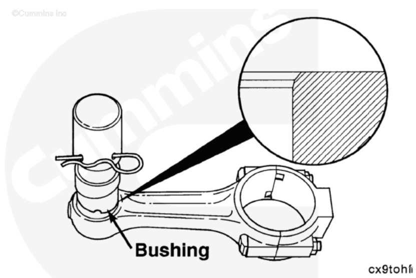
Install the cap and secure with the hair pin cotter pin.



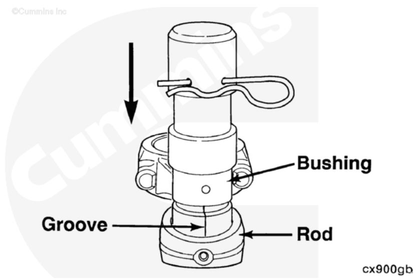
Insert the mandrel and components into the chamfered side of the connecting rod.

Align the angles of the bushing, connecting rod, and stop rings.

Note : Not all connecting rods have a double chamfer. A chamfer can be added to aid in the installation of the bushing, but must not result in an unsupported bushing area.



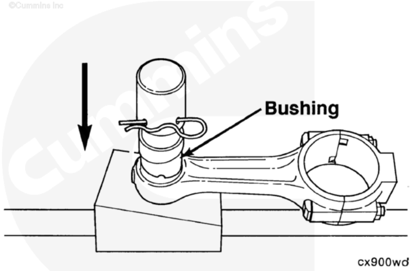
Align the oil hole in the bushing and connecting rod (use the groove in the mandrel) and push the mandrel and components into the pin bore of the connecting rod until contact is made. This contact helps maintain the alignment of the oil holes.



Place the connecting rod and mandrel assembly on the anvil.

Position it so that the angle on the connecting rod matches the angle on the anvil.

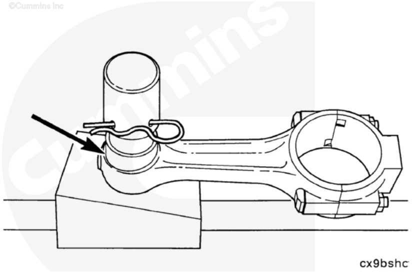
The connecting rod is in the horizontal position.



Press the bushing into the connecting rod by applying force on the top of the cup.

Use either an arbor or hydraulic press.

Push through in a continuous motion until the driver ring makes contact with the connecting rod.



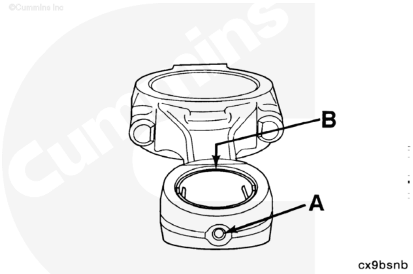
Check the alignment of the oil holes.

A 3.00 mm [0.118 in] diameter rod **must** move freely through the connecting rod and the bushing oil holes (A).

Check for clearance between the connecting rod bushing and the connecting rod.

Use a 0.025 mm [0.001 in] feeler gauge to make sure the bushing is properly seated.

The 0.025 mm [0.001 in] feeler gauge **must not** enter between the bushing and the connecting rod (B).

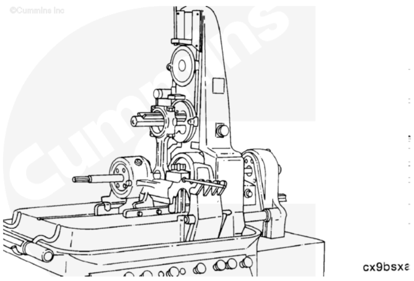


Machine

The inside diameter of new connecting rod bushings **must** be machined with a rod boring machine, such as Part Number 3823601 (Sunnen PM-300®) or Part Number 3375144 (Tobin Arp®).

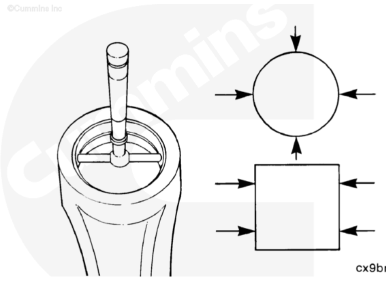
When properly adjusted, a Sunnen PM-300®, or equivalent, connecting rod boring machine is capable of maintaining all the critical dimensions.

Do **not** use a "floating rod" type honing machine.



After machining the connecting rod bushings, it will be necessary to check all critical dimensions.

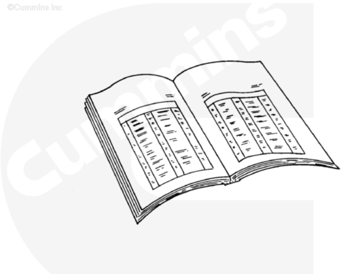
See the instructions within this procedure.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Assemble and install the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/39/39-001-054-tr.html)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-035-tr-gc.html)
- Install the lubricating oil pan and gasket. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-025-tr-gc.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/39/39-002-004-tr-gc.html)
- Fill the lubricating oil pan. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-037.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-018-tr-gc.html)
- Connect the batteries. See equipment manufacturer service information.

- Open the fuel supply valve. See equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for leaks.

Last Modified: 14-Mar-2016